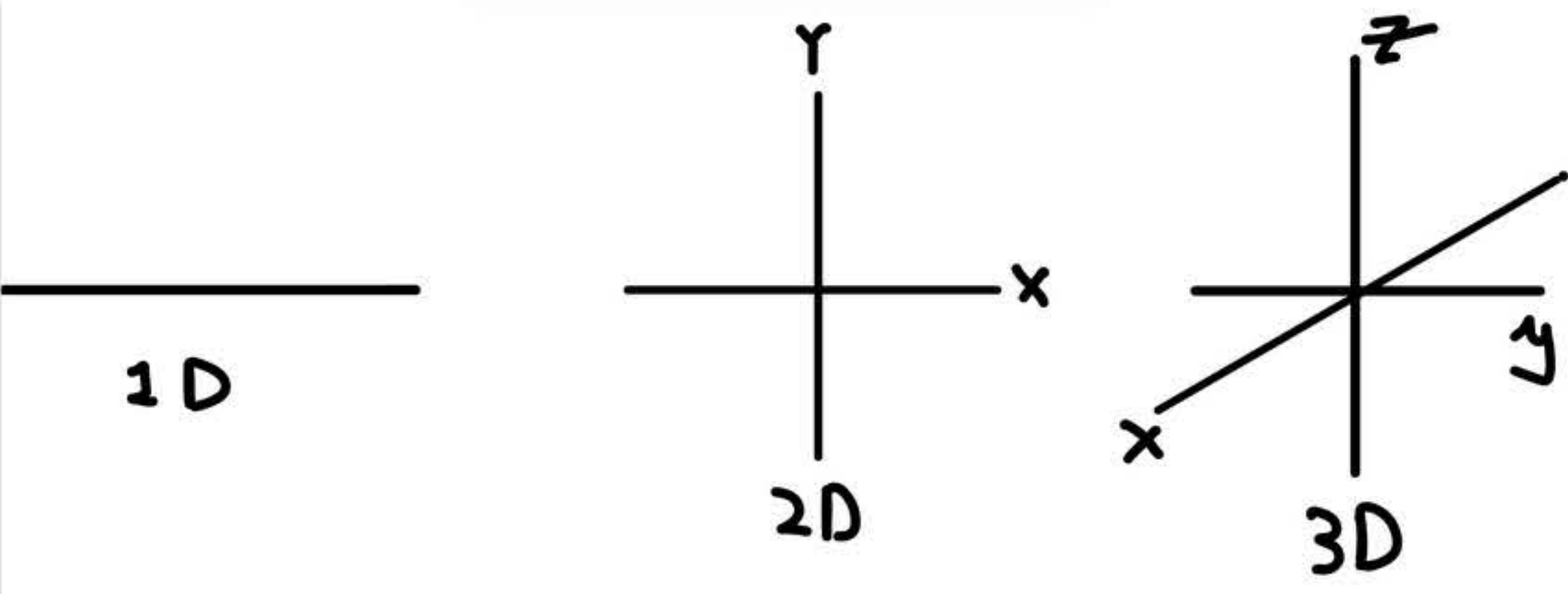
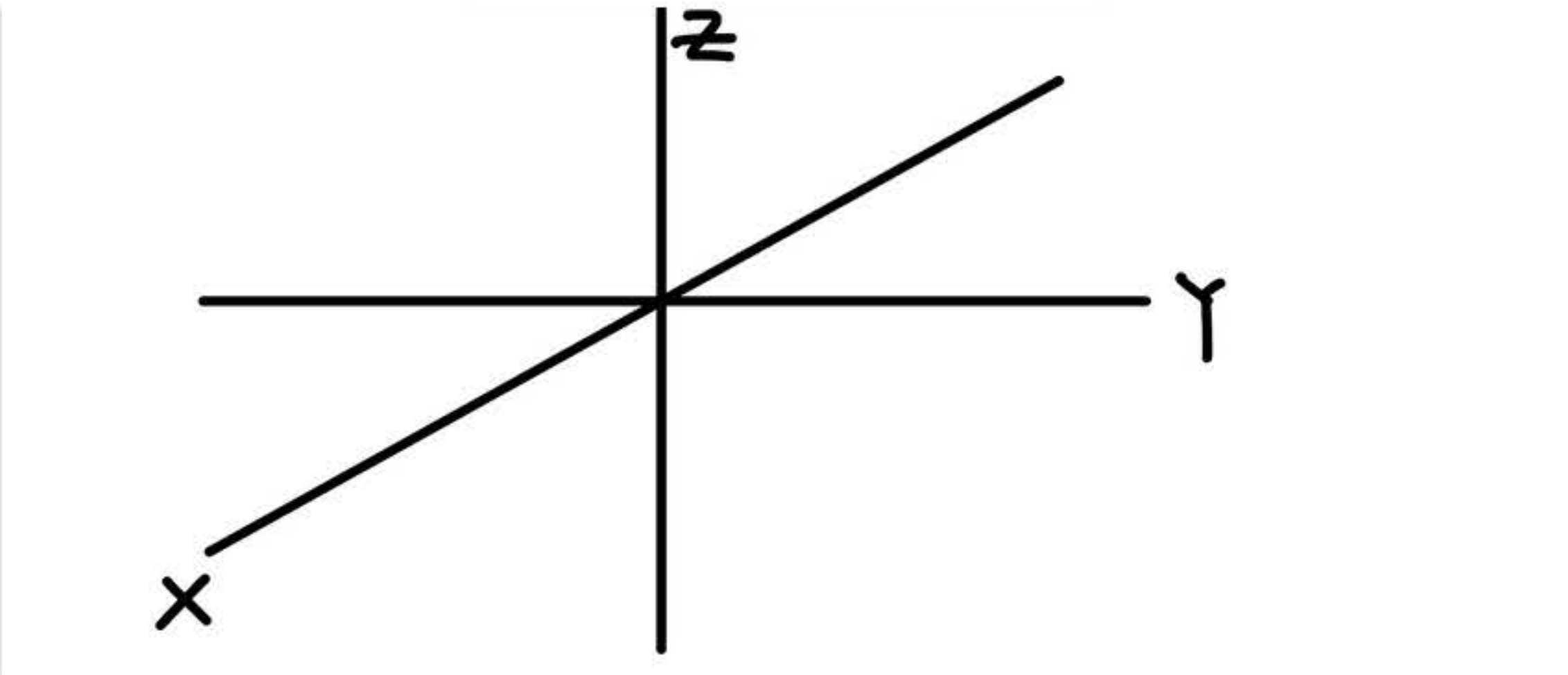


Intro: everything we have before calc3 is in **1D/2D**, in here, we about to change it to **3D**

Here is a diagram I drew to show change of 1D, 2D, and 3D

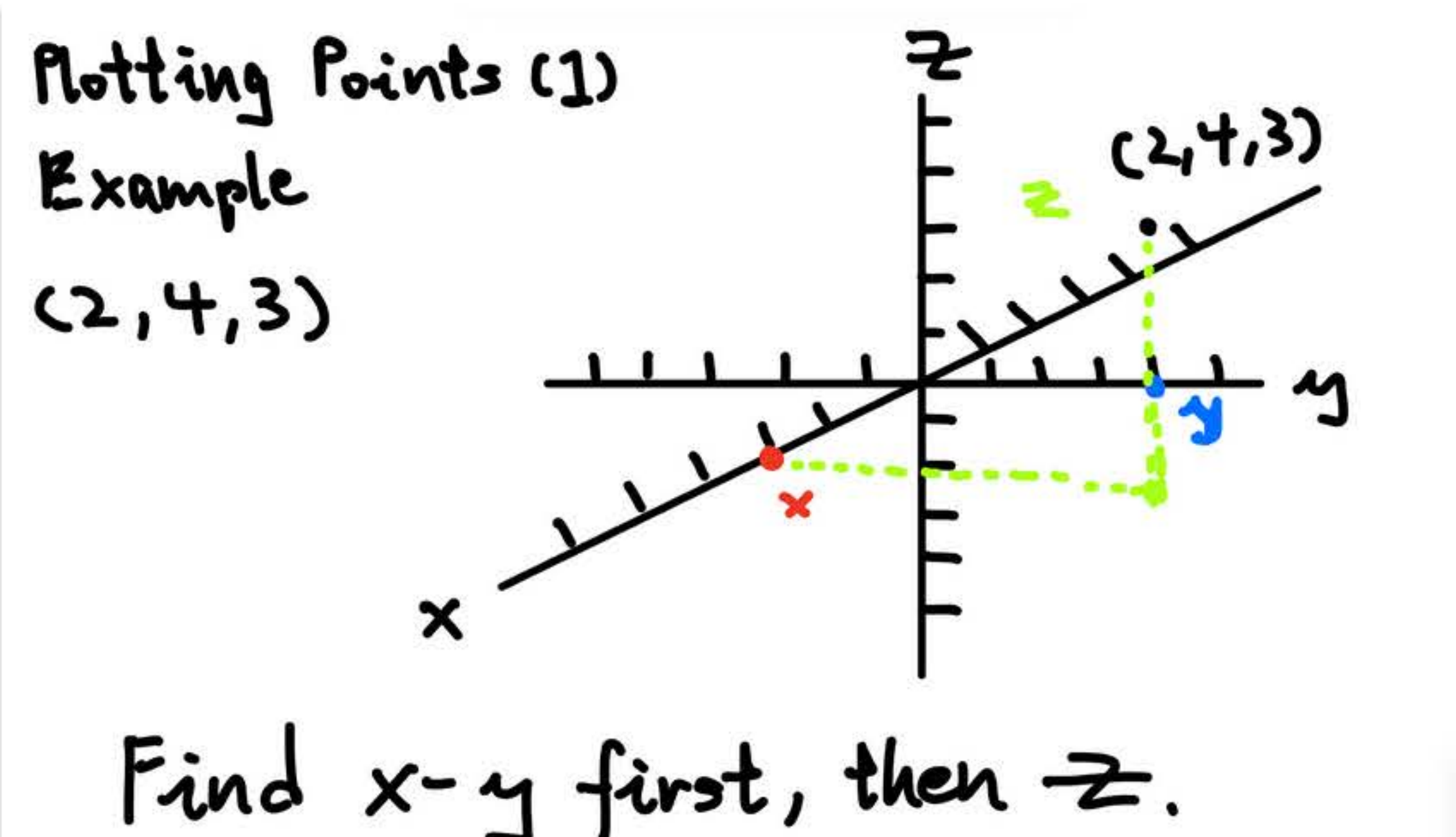


In 3D, the axis(z) are perpendicular with other two axis, on the following image, I'll redraw the diagram of x,y,and z



Plotting Points

To plot a point in 3D, we need x,y,z for coordination.
The idea is we draw the x and y first, then we found z.



Distance Between Points

we use *Perthegram Theorem* to find the distance between two points

Formula:
$$\sqrt{(x2 - x1)^2 + (y2 - y1)^2 + (z2 - z1)^2}$$

Here is Some example problems to work through:

Example: we have points **P** (1,2,3) and points **Q** (8,0,4), how do we find the value of |PQ|?

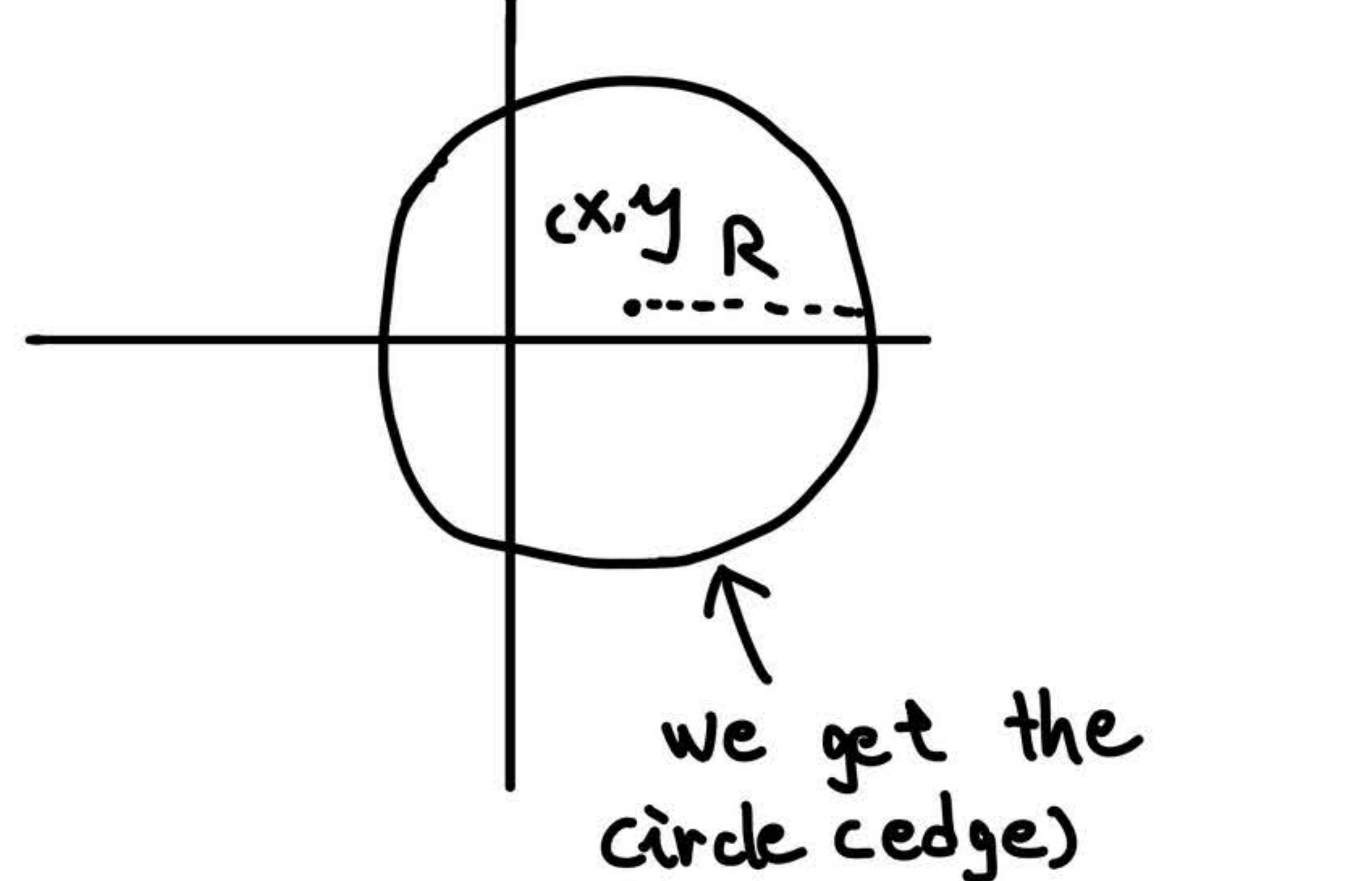
Solution: $|PQ| = \sqrt{(8 - 1)^2 + (0 - 2)^2 + (4 - 3)^2} = \sqrt{7^2 + (-2)^2 + 1^2} = \sqrt{54}$

Circles and Spheres

Recall: In 2D, a circle centered at (x0,y0) with radius R has the equation: $(x - x0)^2 + (y - y0)^2 = R^2$

(The circle we get is the edge part)

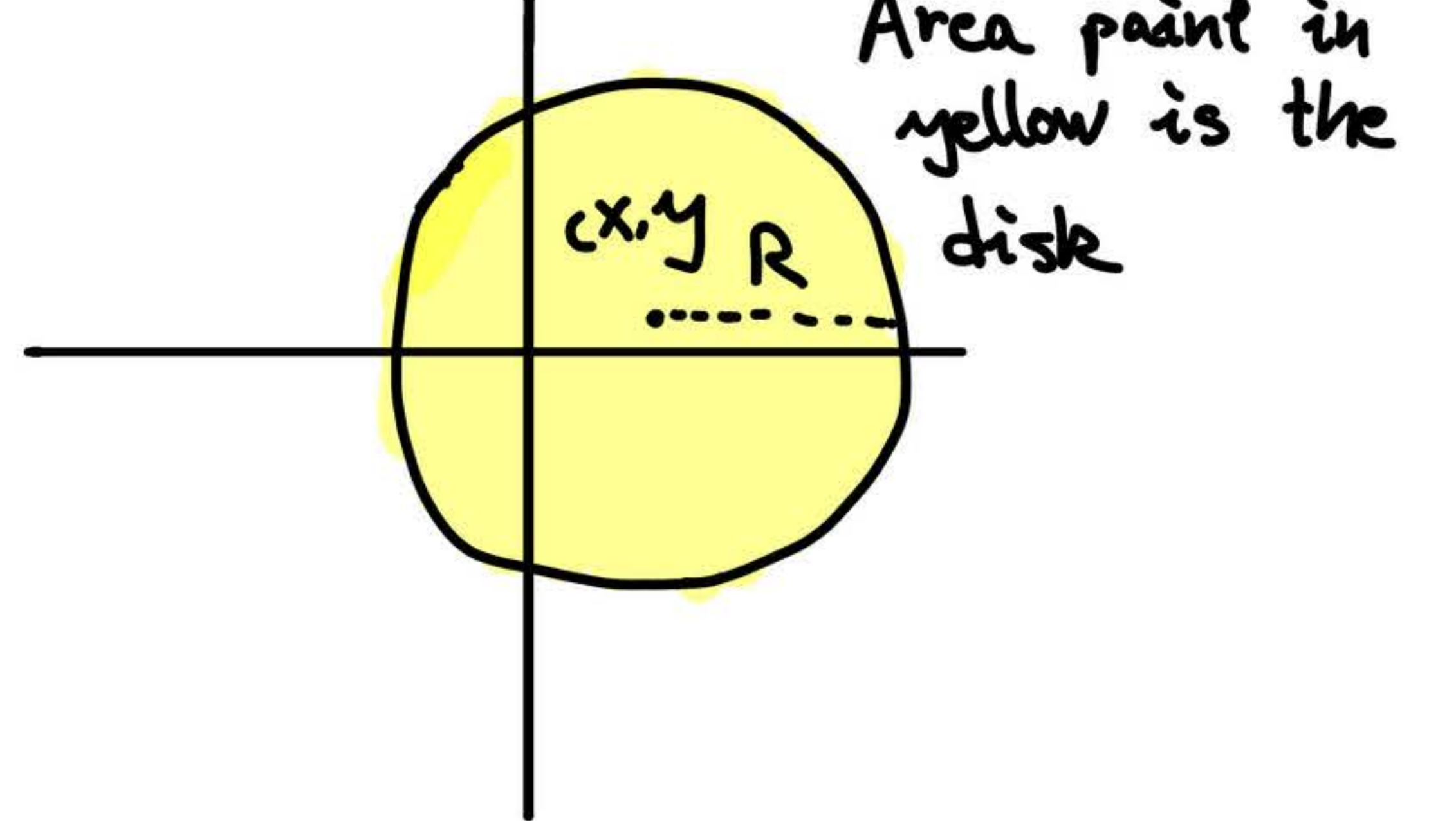
Diagram:



Also we can get disk from an inequality:

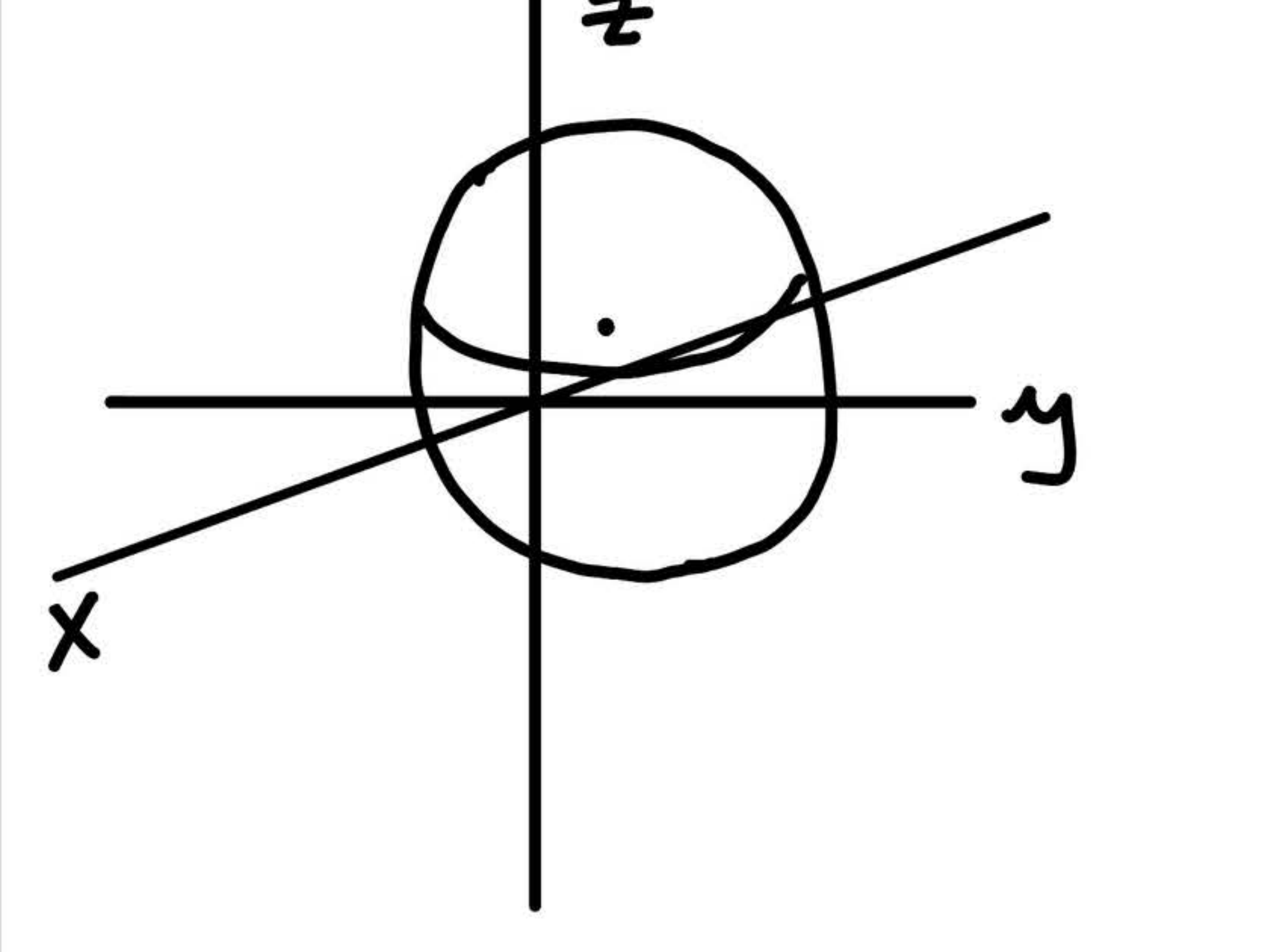
Formula: $(x - x0)^2 + (y - y0)^2 \leq R^2$

Diagram:



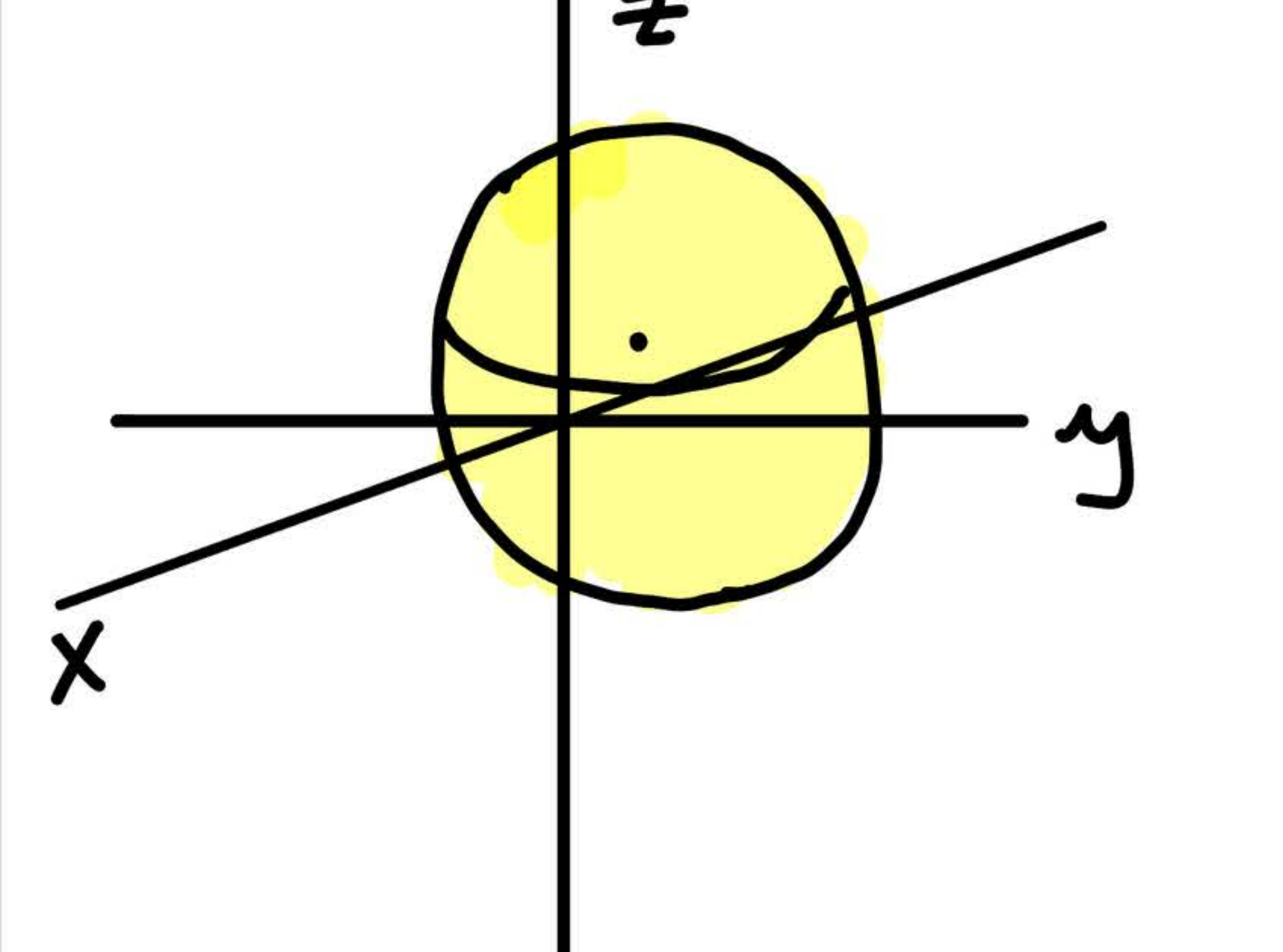
In **3D** a sphere centered at (x0,y0,z0) the radius R has equation: $(x - x0)^2 + (y - y0)^2 + (z - z0)^2 = R^2$

Diagram:



And the equation for ball is: $(x - x0)^2 + (y - y0)^2 + (z - z0)^2 \leq R^2$

Diagram:



Important

In calc1, the equation $y + x^2 = 5$ is a parabola and it's easy to find because it's same as:
 $y = -x^2 + 5$

But the equation of a sphere may not be immediately obvious.